

Introduction to dosage forms and routes of drug administration

By:- Yonas B.

(B. Pharm, MSc)

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- ❑ Principles of dosage form design
- ❑ The Need for Dosage Forms
- ❑ Types of dosage form
- ❑ Pharmaceutical ingredients
- ❑ Routes of drug administration

Objectives

- ❑ After the end of this topic, students will be able to :
 - ❑ Understand the principal objective of dosage form design
 - ❑ Discuss the Need for Dosage Forms
 - ❑ Classify the various types of dosage forms
 - ❑ Describe the importance of pharmaceutical ingredients used
 - ❑ Identify the different Routes of drug administration

Introduction

What is Pharmaceutics?

- **Pharmaceutics** - the area of study concerned with the formulation, manufacture, stability, and effectiveness of dosage forms.
- The *art* and *applied science* of dosage form design
- Pharmaceutics is a broad field of all subject areas in pharmaceutical sciences and encompasses many disciplines
 - Physical pharmaceutics
 - Understanding of basic physical chemistry necessary for the efficient design of dosage form

Introduction, cont'd

- Dosage form design
 - The design and formulation of medicines.
- The manufacture of these medicines
 - Small scale (compounding)
 - Large scale (pharmaceutical technology)
- Microbiology:??
 - the cultivation, avoidance and elimination of MO in medicines

Introduction, cont'd

- **Physical pharmacy??**
- is the subject that deals with the physicochemical principles involved in drug manufacture
- It attempt to integrate the factual knowledge of pharmacy through the development of broad principles of its own and aids the pharmacist in its attempt to predict:
 - ✓ solubility,
 - ✓ compatibility,
 - ✓ stability,
 - ✓ rate of absorption,
 - ✓ duration of action, etc...of drug products.

Introduction, cont'd

- It enables pharmaceutical scientist
 - ✓ to develop new drugs and dosage forms, and
 - ✓ to improve upon various modes of administration (sustained, immediate release,)
- Generally, the subject helps to design an effective, safe and stable formulation.

Principles of dosage form design

- Dosage form is the *physical form of a dose* of a chemical compound used as a drug or medication intended for administration
- Drugs are rarely administered as pure chemical substances alone
 - Used as formulated preparations using appropriate additives or excipients in the formulations
- The excipients provide varied and specialized pharmaceutical functions

Principles, cont'd

- The principal objective of dosage form design
 - To achieve a predictable therapeutic response to a drug included in a formulation with reproducible product quality
- To ensure product quality, numerous features are required
 - ✓ Chemical and physical stability
 - ✓ Suitable preservation against microbial contamination
 - ✓ Uniformity of dose of drug
 - ✓ Acceptability to users
 - ✓ Suitable packaging and labeling

Principles, cont'd

- The proper design and formulation of a dosage form requires consideration of the physical, chemical, and biologic characteristics of
 - ✓ all of the drug substances and pharmaceutical ingredients to be used in fabricating the product

Principles, cont'd

- Before a drug substance is successfully formulated into a dosage form many factors must be considered in order to have high quality and efficacious medicines
 - Biopharmaceutical considerations
 - Drug factors (physicochemical properties of the drug)
 - Therapeutic considerations

The Need for Dosage Forms

- Dosage forms are the means for providing the mechanism
 - For the safe and convenient delivery of accurate dosage
 - Especially drugs of minute or low dose or potent
 - ✓ using diluents
 - ✓ E.g. Ethinyl estradiol (0.05 mg)
- To protect the drug from atmospheric oxygen, humidity (coated tablets)
- To protect the drug from gastric acid (enteric coated)

The Need for Dosage Forms (cont'd)

- To conceal bitter, salty, or offensive taste or odor (capsules., coated tablets, flavored syrups)
- To provide liquid preparations of insoluble drugs or unstable in the desired vehicle (Suspensions)
- To provide clear liquid dosage forms (solutions, syrups)
- To provide rate-controlled drug action (controlled release tablets, capsules, suspensions)

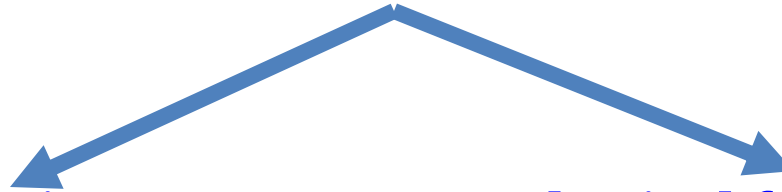
The Need for Dosage Forms (cont'd)

- To provide optimal topical drug action (ointments, creams, patches, ophthalmic, otic, nasal)
- To provide for insertion into body cavity (Rectal or vaginal suppositories)
- To provide for placement of a drug directly into bloodstream or body tissues (injections)
- To provide for inhalation therapy (inhalation aerosols)

Types of dosage forms:

- They are classified according to
 - Route of administration
 - Physical form

Types of dosage forms (Cont'd)



Route of administration

Oral
Topical
Rectal
Parenteral
Vaginal
Inhaled
Ophthalmic
Otic

Physical form

Solid
Semisolid
liquid

Oral dosage forms

Tablet



- ❑ Tablet is a hard, compressed medication in round, oval or square shape
- ❑ The excipients include:
 - ❑ Binders to cause adhesion of the powdered drug and pharmaceutical substances.
 - ❑ glidants improve the flow properties of the powdered drug and granule.
 - ❑ lubricants to ensure efficient tableting

Tablet ,Cont'd

- ❑ Disintegrants to ensure that the tablet breaks up in the digestive tract
- ❑ Sweeteners or flavours to mask the taste of bad-tasting active ingredients.
- ❑ Pigments to make tablets visually attractive

Tablet ,Cont'd

A coating may be applied to:

- 1- hide the taste of the tablet's components
- 2- make the tablet smoother and easier to swallow
- 3- make it more resistant to the external environment extending its shelf life
- 4- sustain drug release
- 5- resist against GI environment??



Buccal and sublingual tablet

- ❑ Sublingual and buccal medications are administered by placing them in the mouth, either under the tongue (sublingual) or between the gum and the cheek (buccal)
- ❑ The medications dissolve rapidly and are absorbed through the mucous membranes of the mouth, where they enter into the bloodstream

Buccal and sublingual tablet, cont'd

- Avoid the acid and enzymatic environment of the stomach and the drug metabolizing enzymes of the liver
- Examples of drugs administered by this route: vasodilator
e.g. Nitroglycerin Sublingual Tablet

Effervescent tablet



- **Effervescent tablets** are uncoated tablets that generally contain acid substances (citric and tartaric acids) and carbonates or bicarbonates and which react rapidly in the presence of water releasing carbon dioxide
- They are intended to be dissolved or dispersed in water before use, providing:
 - very rapid tablet dissolution and dispersion
 - pleasant tasting carbonated drink

Chewable tablet



- They are tablets that chewed prior to swallowing.
- They are designed for administration to children e.g. antacid products

Capsule

A capsule is solid dosage form in which the drug is enclosed within a hard or soft soluble container or shell

The two main types of capsules are:

1- Hard-shelled capsules,

which are normally used for dry, powdered ingredients

2- Soft-shelled capsules primarily used for oils and for active ingredients that are dissolved or suspended in oil.

Hard gelatin capsule



Soft gelatin capsule



Lozenge

- ❑ It is a solid preparation consisting of *sugar* and *gum*, the latter giving strength and cohesiveness to the lozenge and facilitating slow release of the medicament
- ❑ used to medicate the mouth and throat and for the slow administration of indigestion or cough remedies.



Pastilles

- ❑ They are solid medicated preparations designed to dissolve slowly in the mouth
- ❑ They are softer than lozenges and their bases are either glycerol and gelatin, or acacia and sugar



Dental Cones

- A tablet form intended to be placed in the empty socket following a tooth extraction, for preventing the local multiplication of pathogenic bacteria associated with tooth extractions
- The cones contain an antibiotic

Granules

- ❑ They are solid dosage form consisting of dry aggregates of powder particles often supplied in single-dose sachets
- ❑ Some granules are placed on the tongue and swallowed with water, others are intended to be dissolved in water before taking
- ❑ Effervescent granules evolve carbon dioxide when added to water

Powder (Oral)



- ❑ There are two kinds of powder intended for internal use

1- *Bulk Powders* are multidose preparations consisting of solid, loose, dry particles of varying degrees of fineness

- ❑ They contain one or more active ingredients, with or without excipients and, if necessary, coloring matter and flavoring substances

Powder (Oral), cont'd

- ❑ usually contain non-potent medicaments such as antacids.
- ❑ The powder is usually dispersed in water or dissolved before taking

2-Divided Powders are *single-dose* presentations of powder

(e.g. a small sachet) that are intended to be issued to the patient as such, to be taken in or with water.

Liquid preparations

Oral solution

- Oral solutions are clear liquid preparations for oral use containing one or more active ingredients dissolved in a suitable vehicle.

Oral emulsion

- Oral emulsions are stabilized oil-in-water dispersions, either or both phases of which may contain dissolved solids.

Liquid preparations Cont'd

Oral suspension

- Oral suspensions are liquid preparations for oral use containing one or more active ingredients suspended in a suitable vehicle
- Oral suspensions may show a sediment which is readily dispersed on shaking to give a uniform suspension which remains sufficiently stable to enable the correct dose to be delivered

Excipients for Suspension Products

- ❑ In Liquid / suspension products, the possible types of excipients include:
 - ❑ Solvents/co-solvents: A medium in which ingredients of a dosage form are dissolved or dispersed. e.g. Aqueous Vehicle, Propylene Glycol, Glycerol
 - ❑ Buffering agents adjusts the pH of a solution, e.g. Citrate, Gluconates, Lactates
 - ❑ Preservatives: Added to prevent microbial growth, e.g. Na Benzoate, Parabens (Me, Pr and Bu), benzalkonium chloride (BKC)

Excipients for Suspension Products...

- ❑ Anti-oxidants are used to control oxidation of,
 - ✓ API, *e.g.* lovastatin
 - ✓ Preservative, *e.g.* potassium sorbate
 - ✓ Vehicle like oils or fats susceptible to oxidation

e.g. butylated hydroxytoluene (BHT),

Butylated hydroxyanisole (BHA),

Ascorbic acid

Excipients for Suspension Products...

- ❑ Wetting agents are used to aid ‘wetting’ and dispersion of a hydrophobic API, preservative or antioxidant. e.g. Polysorbates, Sorbitan esters
- ❑ Anti-foaming agents are effective at discouraging the formation of stable foams by lowering surface tension and cohesive binding of the liquid phase, e.g. Dimethicone
- ❑ Thickening agents work by entrapment of solid particles in a viscous or ‘gel-like’ structure to prevent settling/sedimentation, e.g. Methylcellulose or Hydroxyethylcellulose

Excipients for Suspension Products...

- ❑ Sweetening agents are used to make the product more palatable and to mask unwanted taste. e.g. Sorbitol, Saccharin, Aspartame, etc.
- ❑ Flavouring agents supplement and complement a sweetening agent, e.g. Peppermint, Lemon oils, etc.
- ❑ Humectants: Their function is to retard evaporation of aqueous vehicle from the dosage form , e.g. Propylene Glycol, Glycerol, Sorbitol

Liquid preparations Cont'd

Syrup

- It is a concentrated aqueous solution of a sugar, usually sucrose
- Usually used as a flavored vehicle for drugs
- Flavored syrups are a convenient form of
masking disagreeable tastes

Liquid preparations Cont'd

Elixir

- ❑ It is pleasantly flavored, clear hydro alcoholic liquid containing dissolved active ingredients intended for oral use
- ❑ The vehicle may contain a high proportion of ethanol or sucrose together with antimicrobial preservatives which confers the stability of the preparation

Liquid preparations Cont'd

Linctuses

- ❑ Linctuses are sweet, viscous liquid oral preparations that are usually prescribed for the relief of cough
- ❑ They usually contain a high proportion of syrup and glycerol which have a demulcent effect on the membranes of the throat
- ❑ The dose volume is small (5ml) and, to prolong the demulcent action, they should be taken undiluted

Liquid preparations, Cont'd

Oral drops:

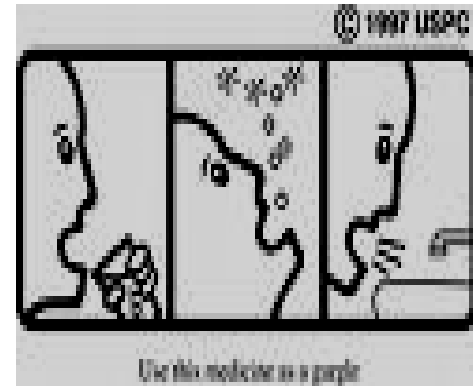
- Oral drops are Liquid preparations for oral use that are intended to be administered in small volumes with the aid of a suitable measuring device
- They may be solutions, suspensions or emulsions



Liquid preparations, Cont'd

Gargles

- ❑ They are aqueous solutions used in the prevention or treatment of throat infections.
- ❑ Usually they are prepared in a concentrated solution with directions for the patient to dilute with warm water before use



Mouthwashes

- ❑ These are similar to gargles but are used for oral hygiene and to treat infections of the mouth.



Topical dosage forms

Ointments:

- ❑ Ointments are semi-solid, greasy preparations for application to the skin, rectum or nasal mucosa
- ❑ The base is usually anhydrous and immiscible with skin secretions
- ❑ Ointments may be used as emollients or to apply suspended or dissolved medicaments to the skin

Topical dosage forms Cont'd

Creams:

- ❑ Creams are semi-solid emulsions

- ❑ They are divided into two types



- A. oil-in-water (O/W) creams:** which are composed of small droplets of oil dispersed in a continuous aqueous phase

- ❑ more comfortable and cosmetically acceptable as they are less greasy and more easily washed off using water.

Topical dosage forms Cont'd

B. water-in-oil (W/O) creams: which are composed of small droplets of water dispersed in a continuous oily phase.

❑ more difficult to handle but many drugs which are incorporated into creams are hydrophobic and will be released more readily from a *W/O* cream than an *O/W* cream.

❑ are also more moisturising as they provide an oily barrier which reduces water loss from the stratum corneum

Topical dosage forms, Cont'd

Gels (Jellies)

- ❑ Gels are semisolid system in which a liquid phase is constrained within a polymeric matrix (consisting of natural or synthetic gum) having a high degree of physical or chemical cross-linking
- ❑ They are used for medication, lubrication and some miscellaneous applications like carrier for spermicidal agents to be used intra-vaginally

Topical dosage forms, Cont'd

Pastes

- ❑ Pastes are basically ointments into which a high percentage of insoluble solid has been added.
- ❑ The extraordinary amount of particulate matter stiffens the system.
- ❑ less penetrating and less macerating than ointment
- ❑ less greasy because of the adsorption of the fluid hydrocarbon fraction to the particulates

Topical dosage forms, Cont'd

- ❑ Pastes make particularly good protective barrier when placed on the skin, the solid they contain can absorb and thereby neutralize certain noxious chemicals before they reach the skin
- ❑ Paste forms an unbroken relatively water-impermeable opaque film
 - ❑ can be used as an effective sun block

Topical dosage forms, Cont'd

Dusting powders

- ☐ very fine powder for external use
- ☐ Not for use on open wounds unless the powders are
sterilized

Topical dosage forms, Cont'd

Transdermal patch

- ❑ a medicated adhesive patch that is placed on the skin to deliver a specific dose of medication through the skin into the bloodstream

- ❑ An advantage of a transdermal drug delivery route
 - ❑ provides a controlled release of the medicament
 - ❑ Bypass first pass metabolism

e.g. scopolamine for motion sickness

Topical dosage forms, Cont'd

Liniments:

- ❑ Liniments are liquid, or semi-liquid preparations intended for rubbing on the body to relieve or soothe pain
- ❑ Most are massaged into the skin (e.g. counter-irritant)
- ❑ They may be alcoholic or oily solutions or emulsions
- ❑ Liniments should not be applied to broken skin.

Topical dosage forms, Cont'd

Lotions

- ❑ These are aqueous suspension or emulsion for external application without friction
- ❑ They are either dabbed on the skin or applied on a suitable dressing and covered with a waterproof dressing to reduce evaporation

Topical dosage forms, Cont'd

Collodion

- ❑ Collodion is a solution of pyroxillon (nitrocellulose) in ether or acetone, sometimes with the addition of alcohols
- ❑ As the solvent evaporates, it dries to a celluloid-like film.
- ❑ Flexible collodion contains pyroxillon (nitrocellulose), castor oil and alcohol in a solvent ether
- ❑ applied to the skin to close small wounds, abrasions, and cuts, and to keep medications in contact with the skin

Topical dosage forms, Cont'd

Paints

- ❑ Paints are liquids for application to the skin or mucous membranes
- ❑ Skin paints contain volatile solvent that evaporates quickly to leave a dry resinous film of medicament
- ❑ Throat paints are more viscous due to a high content of glycerol, designed to prolong contact of the medicament with the affected site

Rectal dosage forms

Suppository:

- ❑ It is a small solid medicated mass, usually cone-shaped, that is inserted either into
 - ❑ the rectum (rectal suppository)
 - ❑ vagina (vaginal suppository or pessaries)
- ❑ melts at body temperature



Rectal dosage forms (Cont.)

Enema is rectal preparation for therapeutics, diagnostics or nutritive purpose

Types of enema:

1 - Evacuation enema (cleansing enema)

- ☐ used as a bowel stimulant to treat constipation.
 - ☐ E.g. soft soap enema & MgSO_4 enema
- ☐ The volume of evacuation enemas may reach up to 2liters.
- ☐ They should be warmed to body T° before administration

Rectal dosage forms, Cont'd

2- Retention enema: The slow infusion of a liquid into the rectum, from where it may or may not be absorbed and not evacuated.

- ❑ their volume does not exceed 100 ml.

- ❑ No warming needed.

- ❑ May exert: local or systemic effect

 - a- Local effect

 - e.g. Barium enema is used as a contrast substance in the radiological imaging of lower bowel

Rectal dosage forms, Cont'd

b- Systemic effect

- ❑ rectal administration of substances into the bloodstream.

- ❑ This may be done in situations where it is impossible to deliver a medication or nutrient by mouth

e.g. nutrient enema which contains carbohydrates, vitamins & minerals

Vaginal dosage forms

Pessary

- ❑ Pessaries are solid medicated preparations designed for insertion into the vagina where they melt or dissolve.
- ❑ There are three types:

1. Moulded pessaries

- ❑ they are cone shaped and prepared in a similar way to moulded suppositories.

Vaginal dosage forms, Cont'd

2. Compressed pessaries

- ❑ made in a variety of shapes and are prepared by compression in a similar manner to oral tablets

3. Vaginal capsules

- ❑ are similar to soft gelatin oral capsules differing only in size and shape

Vaginal dosage forms , Cont'd

Vaginal ring

❑ **Vaginal rings** are 'doughnut-shaped' polymeric drug delivery devices designed to provide controlled release of drugs to the vagina over extended periods of time.

❑ Several vaginal ring products are currently available



Vaginal dosage forms, Cont'd

Douche

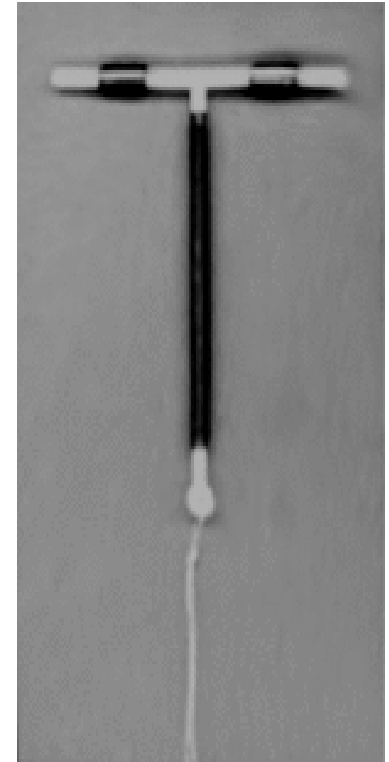
- ❑ Is liquid preparation intended for the irrigative cleaning of the vagina; prepared from powders, liquid solutions or liquid concentrates
- ❑ Douche is a device used to introduce a stream of liquid preparation into the body for medical or hygienic reasons.



Vaginal dosage forms, Cont'd

Intrauterine device (IUD)

- ❑ It is device inserted and left in the uterus to prevent effective conception
- ❑ The IUD is the world's most widely used method of reversible birth control
- ❑ The device(IUD) has to be fitted inside or removed from the uterus by skilled professional
- ❑ Depending on the type, a single IUD is approved for 5 to 10 years use



Inhalation dosage forms

Inhaler

- ❑ Inhalers are solutions, suspensions or emulsion of drugs in a mixture of inert propellants held under pressure in an aerosol dispenser.
- ❑ Release of a dose of the medicament in the form of fine droplets from the container through a spring-loaded valve incorporating a metering device.

Inhalation dosage forms, cont'd

- ❑ The patient then inhales the released drug through a mouthpiece.
- ❑ the valve is actuated by
 - ✓ finger pressure or
 - ✓ patient breathing in through the mouthpiece
- ❑ commonly used to treat asthma and other respiratory problems.

Pressurized dispensers (aerosol sprays):

- ❑ Several different types of pharmaceutical product may be packaged in pressurized dispensers, known as aerosols.
- ❑ Surface sprays produce droplets of 100 μm diameter or greater



Inhalation dosage forms, cont'd

Inhalations

- ❑ Are liquid preparation containing volatile ingredients
- ❑ They may be placed on a pad or added to hot water and vapours inhaled for five to ten minutes
- ❑ Used for treatment of nasal congestion and inflammation of respiratory tract

Ophthalmic dosage forms

Eye drops:

- ☐ **Eye drops** are saline-containing drops used as a vehicle to administer medication in the eye.
- ☐ may contain steroids, antihistamines or topical anesthetics.
- ☐ May only lubricating and tear-replacing solutions.



Ophthalmic ointment & gel:

- ☐ These are sterile semi-solid preparations intended for application to the conjunctiva or eyelid margin.



Otic dosage forms

Ear drops:

- ❑ Ear drops are solutions, suspensions or emulsions of drugs that are instilled into the ear with a dropper.
- ❑ It is used to treat or prevent ear infections, especially infections of the outer ear and ear canal.



Nasal dosage forms

Nasal Drops and Sprays

- ☐ Drugs in solution may be instilled into the nose from a dropper or from a plastic squeeze bottle.
- ☐ The drug may have a local effect
- ☐ The drug may be absorbed through the nasal mucosa to exert a systemic effect.
- ☐ The use of oily nasal drops should be avoided because of possible damage to the cilia of the nasal mucosa

Miscellaneous products

Extracts:

- ❑ These are concentrated preparations containing the active principals of vegetable or animal drugs which have been extracted with suitable solvents and concentrated to form liquid, soft or dry extract.

Tinctures:

- ❑ These are alcoholic or hydroalcoholic preparations containing the active principals of vegetable drugs.
- ❑ They are relatively weak compared to extracts

Miscellaneous products...

Infusions

- ❑ These are dilute solutions containing the readily soluble constituents of crude drugs and prepared by diluting 1 part of concentrated infusion with 9 parts of water
- ❑ Concentrated infusions are prepared by cold extraction of crude drugs with 25% ethanol

Spirits

- ❑ They are alcoholic or aqueous alcoholic solutions of volatile substances used as flavoring agents.

Miscellaneous products...

Aromatic waters

- ❑ These are aqueous solutions, usually saturated with volatile oils or other volatile substances.
- ❑ Used as flavoring agents.

Biopharmaceutical and pharmacokinetic considerations

Biopharmaceutical and pharmacokinetic considerations

- Biopharmaceutics studies the relationship between the physical, chemical and biological sciences applied to drugs, dosage forms and drug action.
- The principles of this subject is important in dosage form design, with regard to drug absorption, distribution, metabolism and excretion

Biopharmaceutical, cont'd

Drug absorption

- In general, a drug must be in solution form before it can be absorbed through the absorbing membranes and epithelia of the skin and GIT and into body fluids
- Medications are made to get through natural body barriers, such as the skin, stomach, intestines, blood-brain barrier, and other membranous tissues
 - ✓ How well the drug passes through these barriers is the one factor that determines its ultimate effectiveness

Biopharmaceutical, cont'd

- Body membranes are generally classified as three main types
 - ✓ those composed of several layers of cells, like the skin;
 - ✓ those composed of a single layer of cells, like the intestinal epithelium; and
 - ✓ those less than one cell thick, like the membrane of a single cell
- The membranes may be viewed in general as a bimolecular lipoid (fat containing) layer attached on both sides to a protein layer

Biopharmaceutical, cont'd

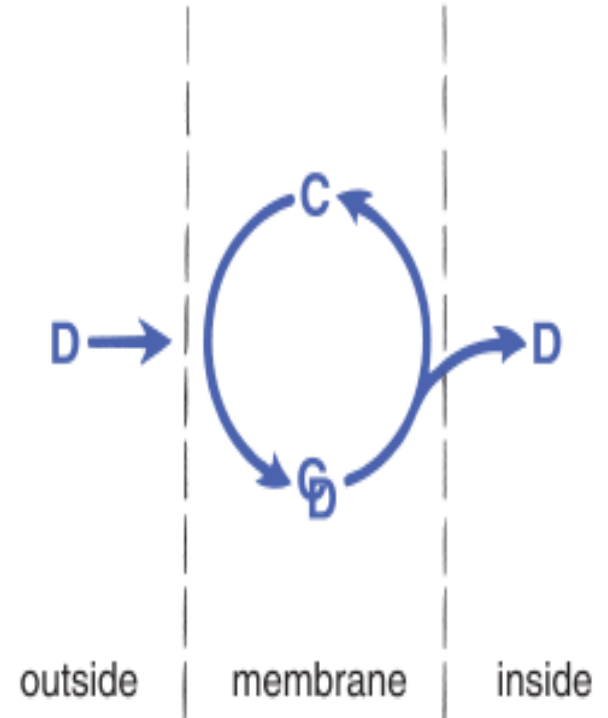
- Drugs are absorbed in two general ways:
 - Passive diffusion
 - Specialized transport systems

■ Passive diffusion

- The term *passive diffusion* is used to describe the passage of (drug) molecules through a membrane that does not actively participate in the process.
- Driven by the concentration gradient that exists across the cellular barrier,
- Controls the absorption of most drugs
- Lipid solubility and degree of ionization of the drug at the absorbing site influence the rate of diffusion

Specialized transport systems

- involves membrane components that may be enzymes or some other type of agent capable of forming a complex with the drug (or other agent) at the surface membrane
- The complex moves across the membrane, where the drug is released, with the carrier returning to the original surface



- Features of specialized transport
 - ✓ process may become saturated as the amount of carrier for a given substance becomes completely bound with that substance
 - ✓ Specificity by a carrier for a particular type of chemical structure
 - ✓ Competitive inhibition if two substances use the same carrier
 - ✓ the transport mechanism is inhibited by general substances that interfere with cell metabolism

Biopharmaceutical, cont'd

❖ Active transport

- Active transport is the movement of a substance against its concentration gradient or electrochemical potential gradient
- Needs energy

❖ Facilitated transport

- The transport of materials is down a concentration gradient, assisted by carrier proteins embedded in the membrane
- No need of energy

Biopharmaceutical, cont'd

- Once absorbed, the drug can exert a therapeutic effect either locally or at the site of action
 - ✓ In the latter case the drug has to be transported in body fluids
- Following absorption; distribution, metabolic and elimination processes will commence immediately

Pharmacokinetics

- **Pharmacokinetics** : The area of study that elucidates the time course of drug concentration in the blood and tissues.
- Considerations such as:
 - Levels of the drug throughout the blood and tissues
 - Overall distribution
 - Metabolism
 - Excretion of the drug
 - Interaction of the drugs with other drugs
 - Life of the drug that includes bioavailability, half-life bioequivalence, and excretion

Distribution

- After the absorption of a medication, it is distributed throughout the body from the bloodstream into tissues
- The distribution of a drug is not necessarily equal throughout the whole body.

Drug Metabolism or Biotransformation

- indicates the chemical changes to drugs within the body as they are metabolized and altered by various biochemical mechanisms
- results in its conversion to one or more compounds that are
 - more water soluble
 - more ionized
 - less capable of binding to proteins of the plasma and tissues
 - less capable of being stored in fat tissue
 - less able to penetrate cell membranes, and
 - less active pharmacologically
- ❖ sometimes the metabolites are more active than the parent compound

Drug Metabolism or Biotransformation

- Four principal chemical reactions are involved in the metabolism of drugs
 - oxidation, reduction, hydrolysis, and conjugation
- Glucuronide conjugation is the most common pathway for drug metabolism, through combination of the drug with glucuronic acid, forming ionized compounds that are easily eliminated via the urine
- There are different factors that influence or alter metabolism such as age, gender, genetics, diet, and other chemicals digested

Excretion

- Excretion is the last phase of a drug's life in the body
- There are many ways a drug can be excreted from the body:
via the kidneys, feces, exhalation, sweat glands, breast milk.
- Urination and bowel movements are the most common
methods of excretion.

Bioavailability

- the rate and extent to which an active drug ingredient or therapeutic moiety is absorbed from a drug product and becomes available at the site of action
- Many drugs travel into the liver before they have a chance to be absorbed into the whole system
 - Known as the “first-pass metabolism”

Half-Life

- Refers to the measurement of the time it takes the body to break down and excrete one half of the drug
- An important factor in the design of dosage form because it tells the manufacturer how long it takes the body to rid itself of the drug

Bioequivalence

- refers to the comparison of bioavailabilities of different formulations, drug products, or batches of the same drug product??
- Generic drug manufacturers strive to achieve the same equivalence to compete with brand name manufacturers.

Routes of Drug Administration

Routes of Drug Administration

- One of the fundamental considerations in dosage form design is whether the drug is intended for local or systemic effects.
- Local effects are achieved by direct application of the drug to the desired site of action
- Systemic effects result from the entrance of the drug into the circulatory system and transport to the cellular site of its action
 - the drug may be placed directly in the blood stream via IV injection or absorbed into the venous circulation following oral or other route of administration

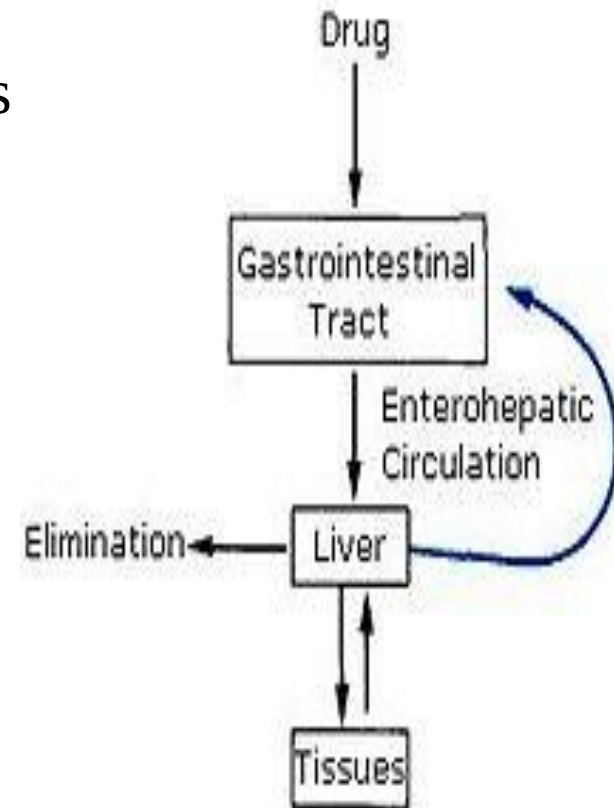
Oral route

- the oral route is considered the most natural, uncomplicated and convenient and safe means of administering drugs.
- Usually intended for systemic effects.
- less expensive,
- the safest means-----retreat is possible unlike parenterals.

Oral route, cont'd

Disadvantages:

- First-pass metabolism
- They do not work as quickly as parenterals
i.e., slow onset of action
 - transit time in GIT,
 - absorption process,
 - enterohepatic blood circulation



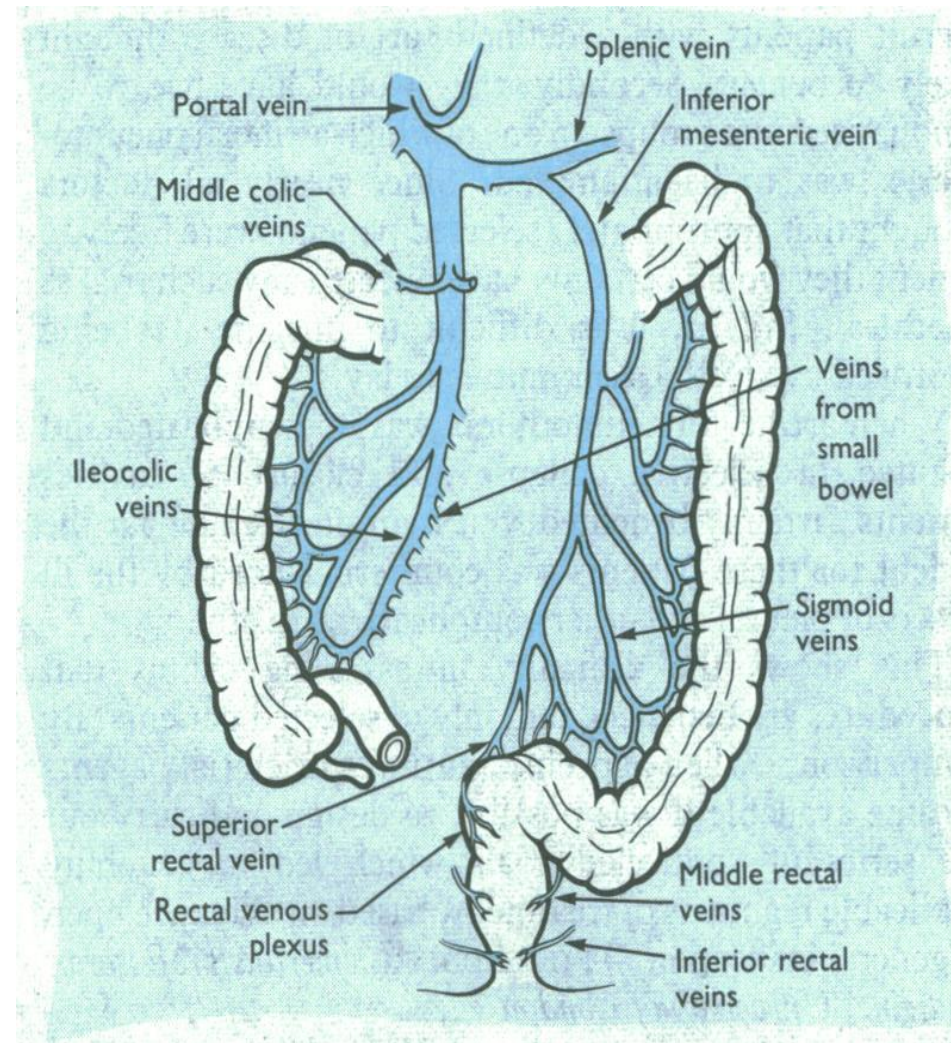
Oral route, cont'd

- chance of irregular absorption of drugs, depending upon such factors as
 - constitutional makeup and the amount or type of food in the gastrointestinal tract
- the destruction of certain drugs by the acid action of the stomach or by gastrointestinal enzymes

- Unsuitable for unconscious or vomiting patients
- Changes in drug solubility due to reactions with other materials,
- Environmental PH can influence the ionization and lipid solubility of drugs

Rectal

- Drugs administered into the rectum are absorbed mainly into the systemic circulation or some may enter into hepatic portal circulation.
- ointments or creams can be used in addition to suppositories



Rectal, cont'd

- For their local effects and systemic effects.
- Rectal route is used:
 - for vomiting patients and unconscious patients,
 - for gastric irritant drugs,
 - for drugs destroyed or inactivated by the environments of the stomach and intestines
- they are uncomfortable
- the actual amount of drug absorbed from rectal mucosa is less predictable than the small intestine

Topical

- Topical preparations are designed to be used for both local or systemic effects.
- May be applied to the skin, placed on the surface of the eye, or used nasally, otically, vaginally, or rectally
- An advantage is easy application.
- A downside is that they might cause a reaction

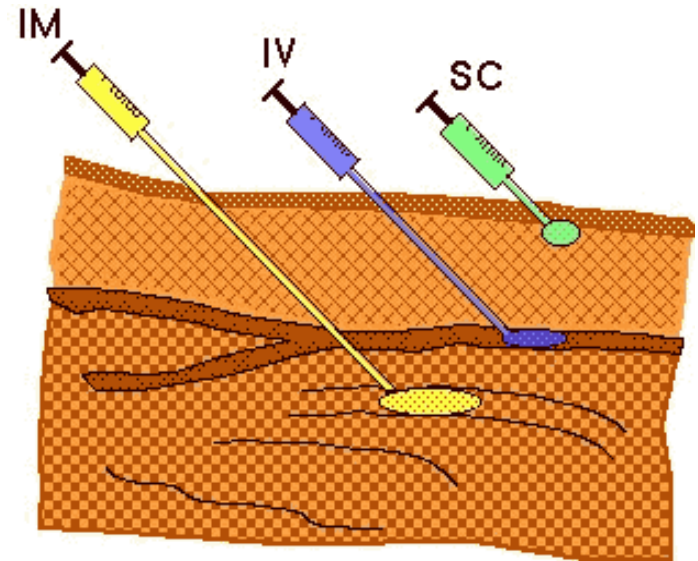
Topical, cont'd

- Topical dermatological product
 - is designed to deliver drug into the skin in treating dermal disorders, with the skin as the target organ.
- Transdermal product
 - is designed to deliver drugs through the skin (percutaneous absorption) to the general circulation for systemic effects, with the skin not being the target organ

Eg, transdermal patches

Parenteral

- Parenteral comes from the Greek
 - *para*, meaning *beside*, and *enteron*, meaning *intestine*
 - means ‘outside of intestine’
- The most common parenteral medications are given IV, IM, SC



Intravenous (IV)

- Drugs may be given into a vein
 - ✓ as a single, small-volume injection (bolus IV injection) or
 - ✓ as a large-volume, slow intravenous drip infusion (slow IV infusion)
- The drug be maintained in solution after injection and not be precipitated within the circulatory system
- oleaginous vehicles are not usually intravenously administered
 - ✓ Intravenous fat emulsion is used for patients receiving parenteral nutrition whose caloric requirements cannot be met by glucose.

Parenteral, cont'd

Advantage

- 100 % bioavailable
- A quick response is possible
- Poorly soluble drugs may be given in a larger volume over an extended period
- useful in emergencies

Disadvantages:

- may be difficult to find a suitable vein
- Requires trained personnel
- Expensive - Sterility, pyrogen testing and larger volume of solvent
- Irreversible effect

Parenteral, cont'd

Intramuscular (IM)

- The injection of a substance deep into the skeletal muscles, generally the gluteal or lumbar muscles
- Aqueous or oleaginous solutions or suspensions may be used IM
- Injection sites are best rotated when a patient is receiving repeated injections over time



Advantages:

- Larger volume (2-5 ml) than SC can be given by IM
- May be easier to administer than IV injections
- Sustained release effect is possible with IM injections, e.g. procaine penicillin

Disadvantage

- Trained personnel required
- Irritating drug may be painful
- Absorption is sometimes erratic, especially for poorly soluble drugs, e.g. diazepam, phenytoin
- The solvent maybe absorbed faster than the drug causing precipitation of the drug at the site of injection

Parenteral, cont'd

Subcutaneous (SC)

- involves administration of the drug by injection just under the skin or in to subcutaneous tissue
- are generally given in the forearm, upper arm, thigh, or buttocks
- If the patient is to receive frequent injections, it is best to alternate injection sites to reduce tissue irritation
- Commonly used for insulin injection

Parenteral, cont'd

- After injection, the drug comes into the immediate vicinity of blood capillaries and permeates them by diffusion
- The capillary wall is an example of a membrane that behaves as a lipid pore barrier
 - lipid-soluble substances penetrating the membrane at rates varying with their oil–water partition coefficients
 - Lipid-insoluble drugs penetrate the capillary membrane at rates that appear to be inversely related to their molecular size

Parenteral, cont'd

- The blood supply to the site of injection is an important factor in considering the rate of drug absorption
 - the closer capillaries are to the site of injection, the more prompt is the drug's entrance into the circulation.
 - the more the capillaries, the more surface area for absorption and the faster the rate of absorption.

Parenteral, cont'd

- Some substances modify the rate of drug absorption from a subcutaneous site of injection
 - The addition of a vasoconstrictor to the injection formulation (or its prior injection) diminish the rate of drug absorption
e.g. the administration of local anesthetics with epinephrine
 - vasodilators may be used to enhance subcutaneous absorption.
- Physical exercise can also influence the absorption of drug from an injection site

Parenteral, cont'd

Advantages:

- Can be given by patient, e.g. in the case of insulin.
- Absorption can be fast from aqueous solution but slower with depot formulations??
- Absorption is usually complete and improved by massage or heat

Disadvantages:

- Can be painful
- Irritant drugs can cause local tissue damage
- Maximum of 2 ml injection

Parenteral, cont'd

Generally, injectable preparations:

- are always sterile
- Benefit: speed of action
- Important
 - for emergency situations
 - for unconscious patients
 - for those who are unable to swallow
 - for drugs inactivated by oral route

Parenteral, cont'd

- More expensive and require preparation and administration by trained personnel??
- Once a drug is injected, there is little time to alter its course if too much drug is given (esp. IV)

Eye, Ear, Nose, cont'd

- Drugs are frequently applied topically to the eye, ear, and mucous membranes of the nose
 - ✓ ointments, suspensions, and solutions
- Eye, ear, and nose preparations usually are not used for systemic effects
 - ✓ ophthalmic and otic preparations are not usually absorbed to any great extent,
 - ✓ nasal preparations may be absorbed, and systemic effects after the intranasal application of solution are fairly common

Eye, Ear, Nose, cont'd

- All eye preparation are sterile.
- Ophthalmic do not last as long as other preparation because of the blinking of the eye and tearing.
- Most nasal preparations are solutions or suspensions administered by drops or as a fine mist
- Ear preparations are usually viscid so that they have prolonged contact with the affected area
 - ✓ Most ear preparation are for clearing up infections or cleaning out ear wax buildup

Inhalation

- The lungs provide an excellent absorbing surface for the administration of gases and for fine aerosol mists (liquids or solids)
- Onset of action is very quick due to rapid absorption through the large surface area of alveolar membrane
- The size and solubility of the drug affect rate of absorption
 - ✓ The size of the drug largely determines the depth to which they penetrate the alveolar regions;
 - ✓ Solubility of drug determine the extent to which they are absorbed

Inhalation

- Soluble drug particles of 0.5 to 1.0 μm size
 - ✓ reach the minute alveolar sacs and are most prompt and efficient in providing systemic effects.
- Particles $< 0.5 \mu\text{m}$
 - ✓ exhaled to some extent and thus, their absorption is not total but variable
- Particles 1 to 10 μm
 - ✓ reach the terminal bronchioles and to some extent the alveolar ducts and provide local effect

Drug factors in dosage form design

- Knowledge of physical and chemical properties of drug substances is important to achieve a stable and effective product.
- These properties such as dissolution rate, solubility, crystal properties, solid state stability, drug-additive interactions can have profound effects on the physiological availability and physical and chemical stability of the drug.

Solubility

- Some aqueous solubility is required for therapeutic efficacy.
- Insoluble compounds exhibit erratic or incomplete absorption thus need use of more soluble salts or other chemical derivatives.

Solubility and Particle Size

- Particle size : determines drug dissolution rate, bioavailability, dosage form content uniformity, flow characteristics, stability such as sedimentation rates
- Particle size reduction:- increase in SA of powders;
- Small increases in solubility can be achieved by particle size reduction.
- Decreases in particle size may enhance dissolution rates

Solubility and pH

- pH can affect solubility.
- pH: solubility profiles are important.

Dissolution

- Dissolution rate is the time it takes for the drug to dissolve in the fluids at the absorption site
- may be rate-limiting step in the absorption of poorly soluble drugs
 - ✓ Can affect onset, intensity, and duration of response and control overall bioavailability of the drug from the dosage form
- The dissolution rate of drugs may be increased by
 - ✓ decreasing the drug's particle size.
 - ✓ increasing its solubility in the diffusion layer.
- The most effective means of obtaining higher dissolution rates is to use a highly water-soluble salt of the parent substance

Partition Coefficient and pKa (Dissociation Constants)

- **pKa (Dissociation Constants)**

- Extent of dissociation or ionization
- The extent of dissociation or ionization in many cases is highly dependent on the pH of the medium containing the drug
- Can affect absorption, distribution, and elimination

Partition Coefficient and pKa cont'd

- For soluble compounds the rate of permeation across biological membranes is the rate determining step
- The permeation rate is dependent upon
 - the size,
 - relative aqueous and lipid solubility (partition coefficient) and
 - degree of ionization of the drug
- Majority of drugs are weak acids or bases,
 - thus depending on their PH, exist in an ionized or unionized forms

Partition Coefficient and pKa, cont'd

- Membranes of absorbing mucosa are more permeable to unionized forms of the drugs
 - greater lipid solubility of unionized forms
 - highly charged nature of cell membranes, which results in the binding or repelling of the ionized drug thereby decreasing penetration

Crystal properties

- Drug substances can be amorphous (without regular molecular lattice arrangement), crystalline, anhydrous, hydrated or solvated
- Many drug substances can exist in more than one form with different molecular packing arrangements in the crystal lattice
 - This property is termed as polymorphism

Crystal properties, cont'd

- Polymorphs are different crystal forms of the same compound, are chemically identical but differ in physical properties
 - dissolution rate, solubility and melting point, as well as powder flow and compaction during tableting

Stability

- Drug substances may degrade or decompose as a result of the effects of heat, oxygen, light, moisture, ...

Eg, esters such as aspirins and procaine are susceptible to hydrolysis.

Organoleptic properties

- Pharmaceutical dosage forms should be acceptable to the patient
- Flavors are used in liquid dosage forms intended for oral administration
- Unpleasant tastes can be overcome by using water-insoluble derivatives of the drugs which have little or no taste. eg, amitriptyline pamoate
- Unpleasant drugs can be administered in capsules or as coated particles or tablets
- Use of Colorants

Therapeutic considerations

- Considerations of the nature of the clinical indication, disease or illness against which the drug is intended and patient factors
- Factors such as :
 - nature of the illness;
 - the manner in which the disease is treated;
 - the need for systemic or local therapy
 - the duration of action required;
 - the age and anticipated condition of the patient.

Therapeutic considerations, cont'd

- Asthmatic patients use inhalation for emergency (b/c, rapidly absorbed) and oral products for chronic therapy
- Children can have difficulty in swallowing solid dosage forms thus many oral preparations are prepared as pleasantly flavored syrups or other liquid dosage form.
- Adults generally prefer solid dosage forms, due to their convenience
- For patients unable to take oral medication or comatose, injectable drugs are used as emergency basis.